

Characteristics Associated with Reported Violations in Regulatory Food Manufacturing Facility Inspections, 2022–2025

Internal analytic summary

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Background

Regulatory inspections are an important component of food manufacturing oversight and can help identify patterns associated with increased likelihood of reported violations. Understanding which facility and inspection characteristics are associated with violations may help support internal monitoring, quality improvement, and future decision-making.

The objective of this analysis was to identify characteristics associated with reported violations in regulatory food manufacturing facility inspections conducted from January 1, 2022 through December 31, 2025. A secondary exploratory analysis examined whether condensed establishment type categories within High- and Medium-risk facilities appeared to contribute disproportionately to observed violation rates. A tertiary exploratory year-over-year summary assessed annual reported violation rates overall and by risk level and explored whether subgroup composition might help contextualize year-to-year patterns.

Methods

Data source and analytic unit

The analytic dataset included regulatory food manufacturing facility inspections conducted between January 1, 2022 and December 31, 2025. Each row represented one regulatory inspection at a food manufacturing facility. The final analytic sample included 1,486 inspections.

Prior to analysis, records outside the scope of interest were removed, including hemp/cosmetics/micro-processing observations and non-permitted firms. In addition, records coded as OAI were recoded as violation = Yes.

Outcome and covariates

The primary outcome was reported violation status, coded as a binary variable (Yes/No). Variables considered for analysis included risk level, region, square footage category, inspection type, and establishment type. A facility identifier was created so that repeated inspections occurring at the same facility could be linked.

Descriptive analysis

Descriptive summaries were generated overall and by violation status. Categorical variables were summarized using counts and percentages. Table 1 describes the distribution of selected inspection characteristics across the analytic sample.

Regression analysis

Because many facilities contributed more than one inspection, a mixed-effects logistic regression model was selected as the primary analytic approach. A random intercept for facility was included to account for within-facility correlation across repeated inspections.

The final primary model included risk level, region, and square footage category as fixed effects, with facility included as a random intercept. Inspection type was evaluated in an earlier model, but sparse categories led to instability and convergence problems, so inspection type was not retained in the final primary model.

Exploratory subgroup analysis

A secondary exploratory analysis was conducted within High- and Medium-risk facilities using a condensed establishment type variable (`est_type`). Because this variable was not structured consistently across all risk levels, it was not included in the primary overall model. Instead, observed violation rates were summarized separately for establishment type categories within High- and Medium-risk facilities

Exploratory year-over-year summary

A tertiary exploratory analysis summarized annual reported violation rates from 2022 through 2025 overall and by risk level. Additional descriptive summaries examined whether year-to-year differences in subgroup composition within High- and Medium-risk establishment types might help contextualize observed annual patterns.

All analyses were conducted in R.

Results

A total of 1,486 regulatory food manufacturing facility inspections conducted between January 1, 2022 and December 31, 2025 were included in the analysis. Overall, violations were reported in 161 inspections (10.8%).

Facility clustering

Repeated inspections within facilities were common. Among 817 unique facilities, 483 (59.1%) appeared once and 334 (40.9%) appeared more than once; the maximum number of inspections contributed by a single facility was 12. This distribution supported use of a mixed-effects model rather than treating all inspections as independent observations.

Primary mixed-effects model

In the final mixed-effects logistic regression model, risk level was the strongest predictor of reported violations. Compared with Low-risk facilities, High-risk facilities had 8.46 times the odds of a reported violation (95% CI: 4.23, 16.94), and Medium-risk facilities had 4.92 times the odds (95% CI: 2.57, 9.41).

Region was also associated with violation status. Relative to Region 253, Region 255 had 6.63 times the odds of a reported violation (95% CI: 3.00, 14.68), and Region 251 had 2.38 times the odds (95% CI: 1.02, 5.58).

Square footage category was also associated with reported violations. Relative to facilities in the 0–1,000 square foot category, inspections in the 1,001–5,000, 5,001–20,000, and 20,001–40,000 square foot categories had significantly higher odds of a reported violation. The largest square-footage categories were not significantly associated with higher odds of violation, suggesting that the relationship between facility size and violations was not simply monotonic from smallest to largest facilities.

Sensitivity analysis excluding N/A square footage

A sensitivity analysis excluding inspections with missing/unknown square footage produced similar results. In the reduced dataset ($n = 1,363$), High-risk facilities had 8.31 times the odds of a reported violation relative to Low-risk facilities (95% CI: 3.88, 17.80), Medium-risk facilities had 4.67 times the odds (95% CI: 2.33, 9.38), and Region 255 remained strongly associated with reported violations (OR: 6.70, 95% CI: 2.82, 15.93). The 1,001–5,000 and 5,001–20,000 square foot categories remained significant, while the 20,001–40,000 square foot category was borderline ($p = 0.060$). These findings suggest that the primary results were robust to exclusion of the N/A square footage category.

Exploratory establishment type analysis

A secondary exploratory analysis examined condensed establishment type categories within High- and Medium-risk facilities.

Within High-risk facilities, the highest observed violation rates were seen in Fish/Seafood (37 violations among 143 inspections, 25.9%), Juice (2/11, 18.2%), Ready-to-Eat (24/175, 13.7%), and Preserved Food (11/85, 12.9%). Because the Juice category had a small sample size, that estimate should be interpreted cautiously.

Within Medium-risk facilities, the highest observed violation rates were seen in Bottled Water (7/20, 35.0%) and Baked Goods (46/277, 16.6%), followed by Preserved Food (2/22, 9.1%), Repackaging (10/112, 8.9%), Beverages/Ice (2/32, 6.3%), and Acid Foods (0/20, 0.0%). Because some Medium-risk establishment types had small counts or zero observed violations, these findings should be interpreted as exploratory.

Exploratory year-over-year summary

A tertiary exploratory year-over-year summary showed overall reported violation rates of 9.7% in 2022, 12.9% in 2023, 10.9% in 2024, and 9.6% in 2025. When stratified by risk level, High-risk inspections had rates of 14.0%, 24.5%, 15.8%, and 16.8%, while Medium-risk inspections had rates of 14.3%, 18.9%, 13.0%, and 10.3%. Low-risk inspections had lower rates overall but increased gradually from 1.9% in 2022 to 5.0% in 2025.

Additional descriptive review suggested that the 2023 peak was not explained solely by annual changes in the distribution of risk categories. Reported violation rates were highest in 2023 within both the High-risk and Medium-risk strata, indicating that the pattern reflected more than a simple shift in overall risk mix. Within High-risk facilities, Fish/Seafood represented a larger share of inspections in 2023 than in surrounding years, and Fish/Seafood also had the highest observed violation rate among High-risk establishment types. Together, these findings suggest that the elevated 2023 rate may have been influenced in part by both a higher within-High-risk violation rate and a larger 2023 share of High-risk Fish/Seafood inspections. A similar composition-based explanation was not evident within Medium-risk facilities.

Table 1. Characteristics of inspections overall and by reported violation status

Characteristic	Total	No violation	Violation
Total inspections	1486	1325 (89.2%)	161 (10.8%)
Risk level			
Low	585	565 (96.6%)	20 (3.4%)
Medium	485	418 (86.2%)	67 (13.8%)
High	416	342 (82.2%)	74 (17.8%)
Region			
251	381	342 (89.8%)	39 (10.2%)
252	173	164 (94.8%)	9 (5.2%)
253	266	253 (95.1%)	13 (4.9%)
254	150	144 (96.0%)	6 (4.0%)
255	325	248 (76.3%)	77 (23.7%)
257	191	174 (91.1%)	17 (8.9%)
Square footage			
0 - 1,000	536	503 (93.8%)	33 (6.2%)
1,001 - 5,000	289	252 (87.2%)	37 (12.8%)
5,001 - 20,000	184	153 (83.2%)	31 (16.8%)
20,001 - 40,000	86	72 (83.7%)	14 (16.3%)
40,001 - 80,000	76	63 (82.9%)	13 (17.1%)
80,001 - 150,000	77	70 (90.9%)	7 (9.1%)
150,001 +	123	113 (91.9%)	10 (8.1%)
N/A	115	99 (86.1%)	16 (13.9%)

Table 2. Mixed-effects logistic regression results

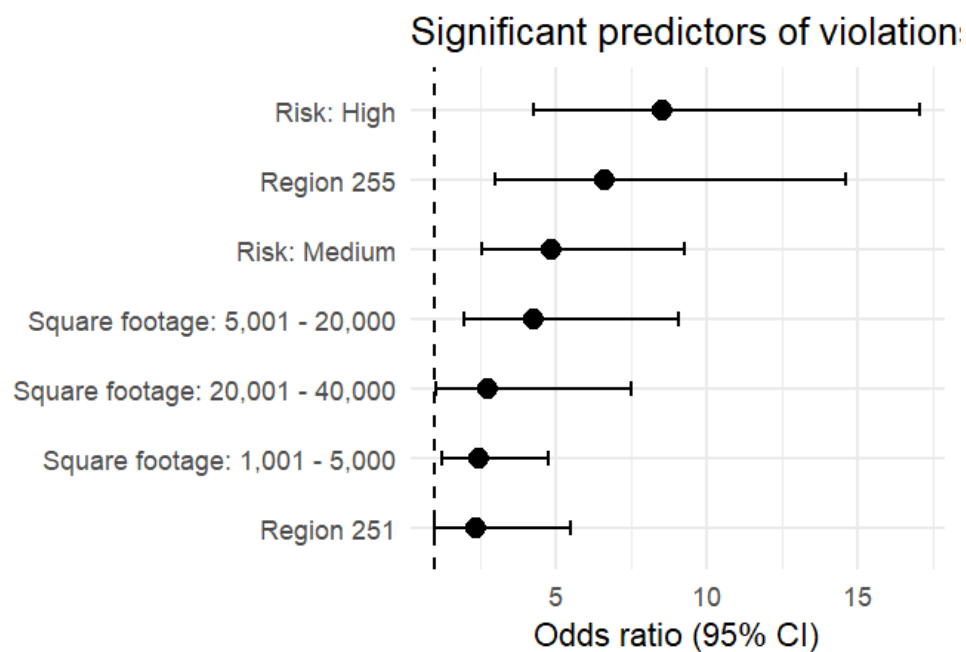
Predictor	Adjusted OR	95% CI	p-value
Risk: High vs Low	8.46	(4.23, 16.94)	<0.001

Predictor	Adjusted OR	95% CI	p-value
Risk: Medium vs Low	4.92	(2.57, 9.41)	<0.001
Region 251 vs 253	2.38	(1.02, 5.58)	0.0460
Region 255 vs 253	6.63	(3.00, 14.68)	<0.001
Square footage: 1,001 - 5,000 vs 0 - 1,000	2.45	(1.25, 4.79)	0.0088
Square footage: 20,001 - 40,000 vs 0 - 1,000	2.76	(1.02, 7.49)	0.0455
Square footage: 5,001 - 20,000 vs 0 - 1,000	4.18	(1.95, 8.96)	0.0002

Table 3. Observed violation rates by condensed establishment type within High- and Medium-risk facilities

Risk group	Establishment type	N	Violations	Proportion with violation
High	Fish_Seafood	143	37	0.259
High	Juice	11	2	0.182
High	Ready_to_Eat	175	24	0.137
High	Preserved Food	85	11	0.129
Medium	Bottled Water	20	7	0.350
Medium	Baked Goods	277	46	0.166
Medium	Preserved Food	22	2	0.091
Medium	Repackaging	112	10	0.089
Medium	Beverages_Ice	32	2	0.063
Medium	Acid Foods	20	0	0.000

Figure 1. Adjusted odds ratios from the final mixed-effects model



*Risk: Low as reference group

*Region 253 as reference group

*Square footage: 0-1000 as reference group

Figure 2. Violation rate by establishment type within High-risk facilities

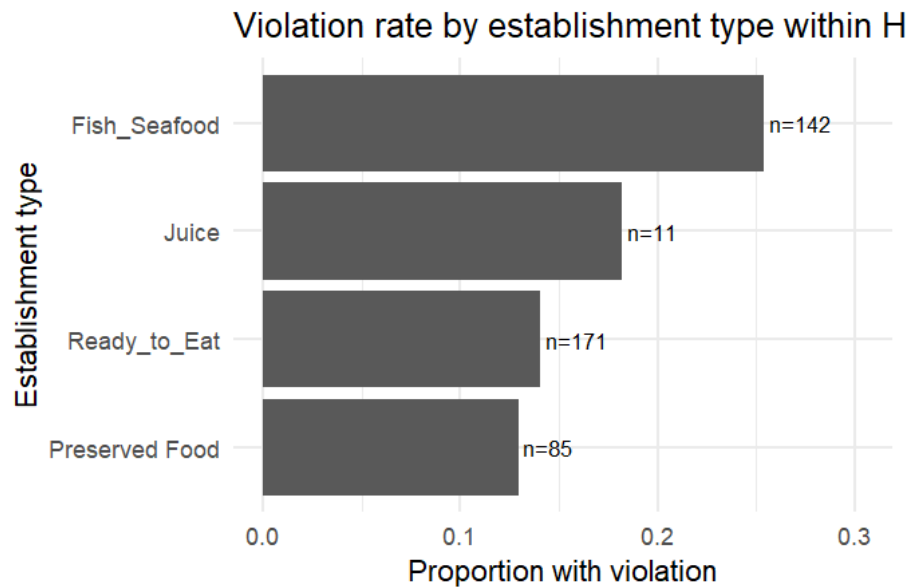


Figure 3. Violation rate by establishment type within Medium-risk facilities

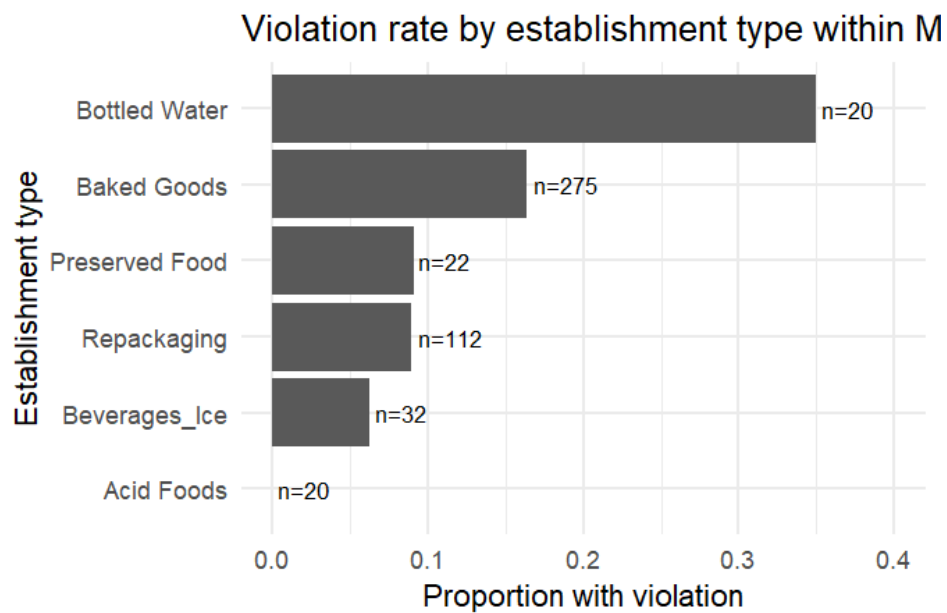
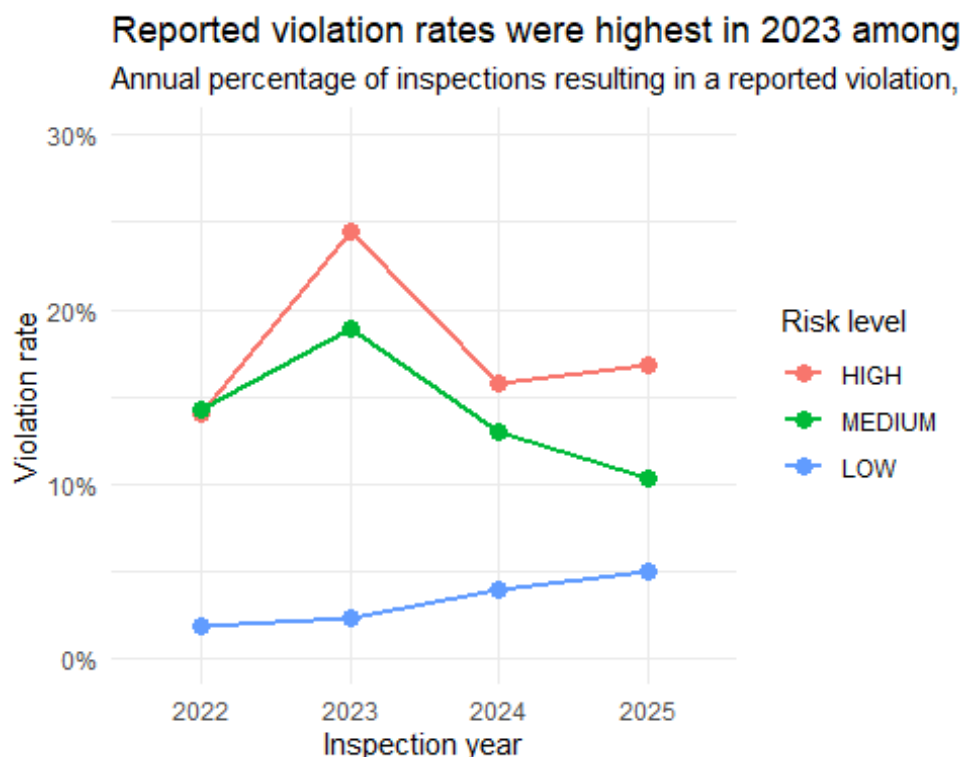


Figure 4. Reported violation rates by year and risk level



Interpretation

This analysis found that risk level was the strongest overall predictor of reported violations in regulatory food manufacturing facility inspections. The association remained strong after accounting for repeated inspections within facilities, suggesting that the relationship was robust to facility-level clustering.

Region and square footage were also associated with reported violations. Several medium-sized square footage categories had higher odds of violation than the smallest facilities, while the largest facilities were not significantly different from the smallest group in the final model. This suggests that the association between facility size and reported violations may be concentrated in mid-sized facilities rather than increasing steadily with facility size.

The exploratory establishment type analysis suggested that elevated violation rates within High- and Medium-risk groups were not uniform across all subtypes. In particular, Fish/Seafood appeared to stand out within High-risk facilities, while Bottled Water and Baked Goods stood out within Medium-risk facilities.

The exploratory year-over-year summary suggested that reported violation rates were highest in 2023, particularly among High- and Medium-risk facilities. Additional descriptive review suggested that this pattern was not explained solely by annual changes in overall risk distribution. Within High-risk facilities, Fish/Seafood represented a larger share of inspections in 2023 than in surrounding years and also had the highest observed violation rate among High-risk establishment types, suggesting that composition of the High-risk subgroup may have contributed in part to the 2023 peak. Because these year-over-year analyses were descriptive, the underlying cause of the 2023 increase could not be determined directly.

Limitations

This analysis has several limitations. First, the dataset was observational and administrative in nature, so the findings should be interpreted as associations rather than causal effects. Second, some variables contained sparse categories, which created instability in more detailed models and limited the ability to retain all candidate

predictors in the final mixed-effects model. Third, the condensed establishment type variable was not structured consistently across all risk levels, so it was used only in exploratory subgroup analyses rather than in the primary overall model. Finally, some establishment type subgroup estimates were based on relatively small sample sizes and should be interpreted cautiously.

Conclusion

In this analysis of regulatory food manufacturing facility inspections from 2022–2025, risk level was the strongest predictor of reported violations, with High- and Medium-risk facilities showing substantially higher odds of violation than Low-risk facilities. Region and square footage were also associated with reported violations in the final mixed-effects model. Exploratory subgroup analyses suggested that specific establishment types within High- and Medium-risk facilities may contribute disproportionately to observed violation patterns, and a year-over-year summary indicated that reported violation rates peaked in 2023, particularly among High- and Medium-risk inspections.

Appendix A. R Markdown output

The following appendix reproduces the rendered R Markdown output used to generate the accompanying analytic tables, figures, and reusable code.

Inspections_Analysis

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2026-06-05

Load appropriate libraries

```
library(readxl)
library(dplyr)
library(janitor)
library(lme4)
library(ggplot2)
library(scales)
```

Load and clean dataset

```
dat <- suppressMessages(
  read_excel("InspectionsAdhocReport.xlsx", sheet = "V2")
) %>%
  clean_names() %>%
  select(-x21, -x164, -x23) %>%
  mutate(
    facility_id = as.factor(facility_id),
    violation = as.factor(violation),
    inspection_begin_date = as.Date(inspection_begin_date, format = "%m/%d/%Y"),
    risk = as.factor(risk),
    square_footage = as.factor(square_footage),
    region = as.factor(region),
    est_type = as.factor(est_type)
  ) %>%
  mutate(
    violation = relevel(violation, ref = "No"),
    risk = relevel(risk, ref = "LOW"),
    square_footage = relevel(square_footage, ref = "0 - 1,000"),
    region = relevel(region, ref = "253"),
    year = format(inspection_begin_date, "%Y"),
    year = as.factor(year)
  )
```

Build mixed-effects logistic-regression model

```
#final mixed-effects logistic regression model
mod_main <- glmer(
  violation ~ risk + region + square_footage + (1 | facility_id),
  family = binomial,
  data = dat,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
)
```



```

summary(mod_main)

##
## Correlation matrix not shown by default, as p = 15 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

#get coefficients and Wald confidence intervals
coef_tab <- summary(mod_main)$coefficients
ci_tab <- confint(mod_main, parm = "beta_", method = "Wald")

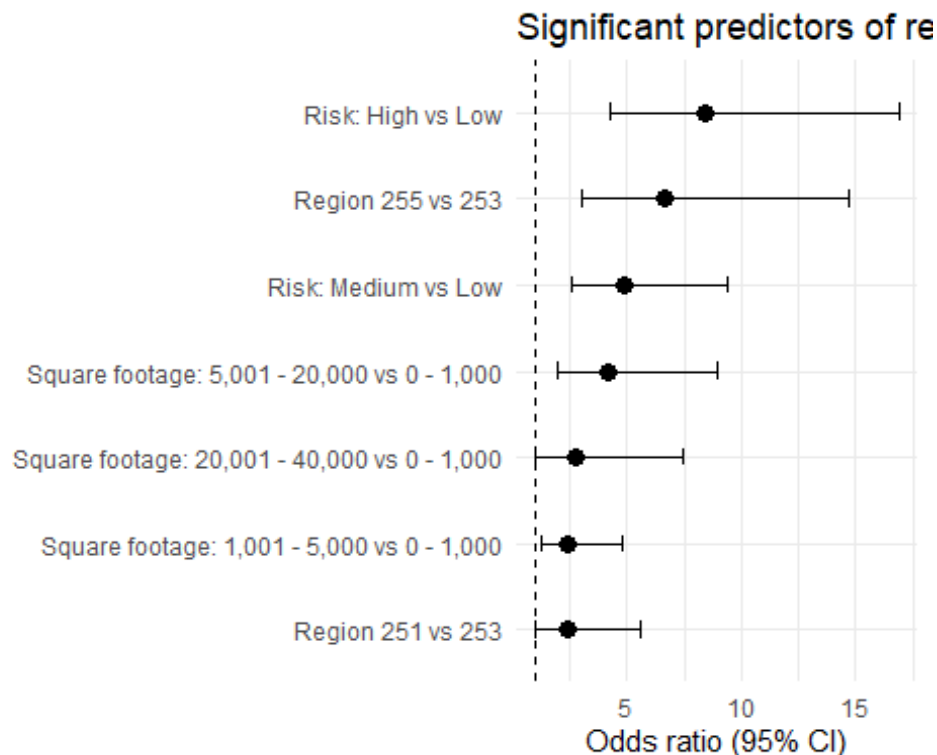
#create main regression table
tbl_main_model <- data.frame(
  term = rownames(coef_tab),
  OR = exp(coef_tab[, "Estimate"]),
  CI_low = exp(ci_tab[, 1]),
  CI_high = exp(ci_tab[, 2]),
  p_value = coef_tab[, "Pr(>|z|)"],
  row.names = NULL
) %>%
  filter(term != "(Intercept)") %>%
  mutate(
    term = recode(
      term,
      "riskHIGH" = "Risk: High vs Low",
      "riskMEDIUM" = "Risk: Medium vs Low",
      "region251" = "Region 251 vs 253",
      "region252" = "Region 252 vs 253",
      "region254" = "Region 254 vs 253",
      "region255" = "Region 255 vs 253",
      "region257" = "Region 257 vs 253",
      "square_footage1,001 - 5,000" = "Square footage: 1,001 - 5,000 vs 0 - 1,000",
      "square_footage150,001 +" = "Square footage: 150,001 + vs 0 - 1,000",
      "square_footage20,001 - 40,000" = "Square footage: 20,001 - 40,000 vs 0 - 1,000",
      "square_footage40,001 - 80,000" = "Square footage: 40,001 - 80,000 vs 0 - 1,000",
      "square_footage5,001 - 20,000" = "Square footage: 5,001 - 20,000 vs 0 - 1,000",
      "square_footage80,001 -150,000" = "Square footage: 80,001 -150,000 vs 0 - 1,000",
      "square_footageN/A" = "Square footage: N/A vs 0 - 1,000"
    ),
    OR = round(OR, 2),
    CI_low = round(CI_low, 2),
    CI_high = round(CI_high, 2),
    p_value = round(p_value, 4),
    `95% CI` = paste0("(", CI_low, ", ", CI_high, ")")
  )

tbl_main_model %>%
  select(term, OR, `95% CI`, p_value)

```

	term	OR	95% CI	p_value
## 1	Risk: High vs Low	8.46	(4.23, 16.94)	0.0000
## 2	Risk: Medium vs Low	4.92	(2.57, 9.41)	0.0000
## 3	Region 251 vs 253	2.38	(1.02, 5.58)	0.0460
## 4	Region 252 vs 253	0.78	(0.24, 2.48)	0.6688
## 5	Region 254 vs 253	1.54	(0.48, 4.92)	0.4655
## 6	Region 255 vs 253	6.63	(3, 14.68)	0.0000
## 7	Region 257 vs 253	2.24	(0.89, 5.66)	0.0870
## 8	Square footage: 1,001 - 5,000 vs 0 - 1,000	2.45	(1.25, 4.79)	0.0088
## 9	Square footage: 150,001 + vs 0 - 1,000	1.57	(0.58, 4.27)	0.3746
## 10	Square footage: 20,001 - 40,000 vs 0 - 1,000	2.76	(1.02, 7.49)	0.0455
## 11	Square footage: 40,001 - 80,000 vs 0 - 1,000	2.68	(0.93, 7.79)	0.0692
## 12	Square footage: 5,001 - 20,000 vs 0 - 1,000	4.18	(1.95, 8.96)	0.0002
## 13	Square footage: 80,001 -150,000 vs 0 - 1,000	1.95	(0.62, 6.08)	0.2526
## 14	Square footage: N/A vs 0 - 1,000	2.22	(0.72, 6.83)	0.1626

```
ggplot(
  tbl_main_model %>% filter(p_value < 0.05),
  aes(x = OR, y = reorder(term, OR))
) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = CI_low, xmax = CI_high), height = 0.2) +
  geom_vline(xintercept = 1, linetype = "dashed") +
  labs(
    title = "Significant predictors of reported violations",
    x = "Odds ratio (95% CI)",
    y = ""
  ) +
  theme_minimal()
```



Descriptive summaries

#overall inspection and violation summary

```
overall_tab <- dat %>%
  summarize(
    total_inspections = n(),
    violations = sum(violation == "Yes"),
    no_violations = sum(violation == "No"),
    pct_violations = round(100 * mean(violation == "Yes"), 1)
  )
```

overall_tab

```
## # A tibble: 1 × 4
##   total_inspections violations no_violations pct_violations
##           <int>       <int>         <int>         <dbl>
## 1             1486         161         1325          10.8
```

#Table 1: risk

```
tab1_risk_wide <- dat %>%
  group_by(risk, violation) %>%
  summarize(n = n(), .groups = "drop") %>%
  tidyr::pivot_wider(names_from = violation, values_from = n, values_fill = 0) %>%
  mutate(
    Total = No + Yes,
    `No violation` = paste0(No, " (", round(100 * No / Total, 1), "%)"),
    `Violation` = paste0(Yes, " (", round(100 * Yes / Total, 1), "%)")
  ) %>%
  select(risk, Total, `No violation`, Violation)
```

tab1_risk_wide

```
## # A tibble: 3 × 4
##   risk   Total `No violation` Violation
##   <fct> <int> <chr>          <chr>
## 1 LOW    585 565 (96.6%)    20 (3.4%)
## 2 HIGH   416 342 (82.2%)    74 (17.8%)
## 3 MEDIUM 485 418 (86.2%)    67 (13.8%)
```

#Table 1: region

```
tab1_region_wide <- dat %>%
  group_by(region, violation) %>%
  summarize(n = n(), .groups = "drop") %>%
  tidyr::pivot_wider(names_from = violation, values_from = n, values_fill = 0) %>%
  mutate(
    Total = No + Yes,
    `No violation` = paste0(No, " (", round(100 * No / Total, 1), "%)"),
    `Violation` = paste0(Yes, " (", round(100 * Yes / Total, 1), "%)")
  ) %>%
  select(region, Total, `No violation`, Violation)
```

tab1_region_wide

```
## # A tibble: 6 × 4
##   region Total `No violation` Violation
##   <fct>   <int> <chr>         <chr>
## 1 253      266 253 (95.1%)    13 (4.9%)
## 2 251      381 342 (89.8%)    39 (10.2%)
## 3 252      173 164 (94.8%)     9 (5.2%)
## 4 254      150 144 (96%)       6 (4%)
## 5 255      325 248 (76.3%)    77 (23.7%)
## 6 257      191 174 (91.1%)    17 (8.9%)
```

#Table 1: square footage

```
tab1_sqft_wide <- dat %>%
  group_by(square_footage, violation) %>%
  summarize(n = n(), .groups = "drop") %>%
  tidyr::pivot_wider(names_from = violation, values_from = n, values_fill = 0) %>%
  mutate(
    Total = No + Yes,
    `No violation` = paste0(No, " (", round(100 * No / Total, 1), "%)"),
    `Violation` = paste0(Yes, " (", round(100 * Yes / Total, 1), "%)")
  ) %>%
  select(square_footage, Total, `No violation`, Violation)
```

tab1_sqft_wide

```
## # A tibble: 8 × 4
##   square_footage Total `No violation` Violation
##   <fct>          <int> <chr>         <chr>
## 1 0 - 1,000      536 503 (93.8%)    33 (6.2%)
## 2 1,001 - 5,000  289 252 (87.2%)    37 (12.8%)
## 3 150,001 +      123 113 (91.9%)    10 (8.1%)
## 4 20,001 - 40,000  86 72 (83.7%)    14 (16.3%)
## 5 40,001 - 80,000  76 63 (82.9%)    13 (17.1%)
## 6 5,001 - 20,000  184 153 (83.2%)    31 (16.8%)
## 7 80,001 -150,000  77 70 (90.9%)     7 (9.1%)
## 8 N/A           115 99 (86.1%)    16 (13.9%)
```

#facility repeat summary

```
facility_freq <- dat %>%
  count(facility_id, name = "n_inspections")

facility_summary <- facility_freq %>%
  summarize(
    n_facilities = n(),
    singletons = sum(n_inspections == 1),
    repeated_2plus = sum(n_inspections >= 2),
    max_inspections = max(n_inspections),
    mean_inspections = round(mean(n_inspections), 2)
  )
```

facility_summary

```
## # A tibble: 1 × 5
##   n_facilities singletons repeated_2plus max_inspections mean_inspections
```

##	<int>	<int>	<int>	<int>	<dbl>
## 1	817	483	334	12	1.82

Exploratory subgroup analysis

#create High- and Medium-risk subsets

```
dat_hm <- dat %>%
  filter(risk %in% c("HIGH", "MEDIUM")) %>%
  filter(est_type != "0") %>%
  droplevels()
```

```
dat_high <- dat_hm %>%
  filter(risk == "HIGH") %>%
  droplevels()
```

```
dat_med <- dat_hm %>%
  filter(risk == "MEDIUM") %>%
  droplevels()
```

High-risk facilities

#summarize establishment type within High-risk facilities

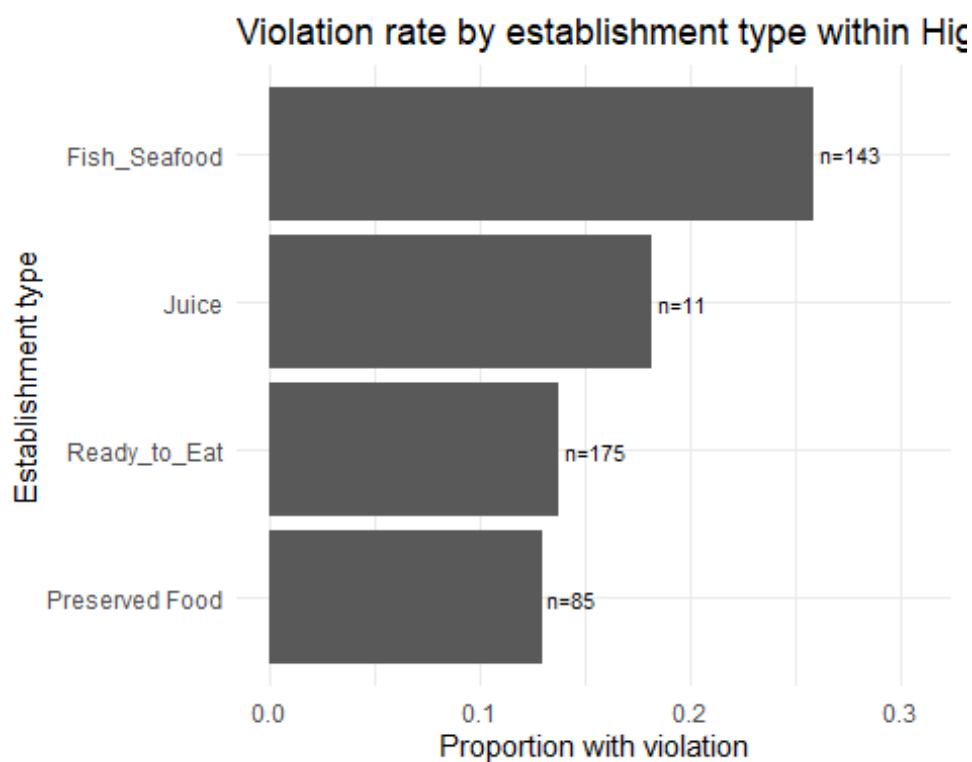
```
tab_high <- dat_high %>%
  group_by(est_type) %>%
  summarize(
    n = n(),
    violations = sum(violation == "Yes"),
    prop_violation = round(mean(violation == "Yes"), 4),
    .groups = "drop"
  ) %>%
  arrange(desc(prop_violation))
```

tab_high

A tibble: 4 × 4

##	est_type	n violations	prop_violation
##	<fct>	<int>	<dbl>
## 1	Fish_Seafood	143	0.259
## 2	Juice	11	0.182
## 3	Ready_to_Eat	175	0.137
## 4	Preserved Food	85	0.129

```
ggplot(tab_high, aes(x = reorder(est_type, prop_violation), y = prop_violation)) +
  geom_col() +
  geom_text(aes(label = paste0("n=", n)), hjust = -0.1, size = 3) +
  coord_flip() +
  expand_limits(y = max(tab_high$prop_violation) + 0.05) +
  labs(
    title = "Violation rate by establishment type within High-risk facilities",
    x = "Establishment type",
    y = "Proportion with violation"
  ) +
  theme_minimal()
```



Medium-risk facilities

#summarize establishment type within Medium-risk facilities

```
tab_med <- dat_med %>%
  group_by(est_type) %>%
  summarize(
    n = n(),
    violations = sum(violation == "Yes"),
    prop_violation = round(mean(violation == "Yes"), 4),
    .groups = "drop"
  ) %>%
  arrange(desc(prop_violation))
```

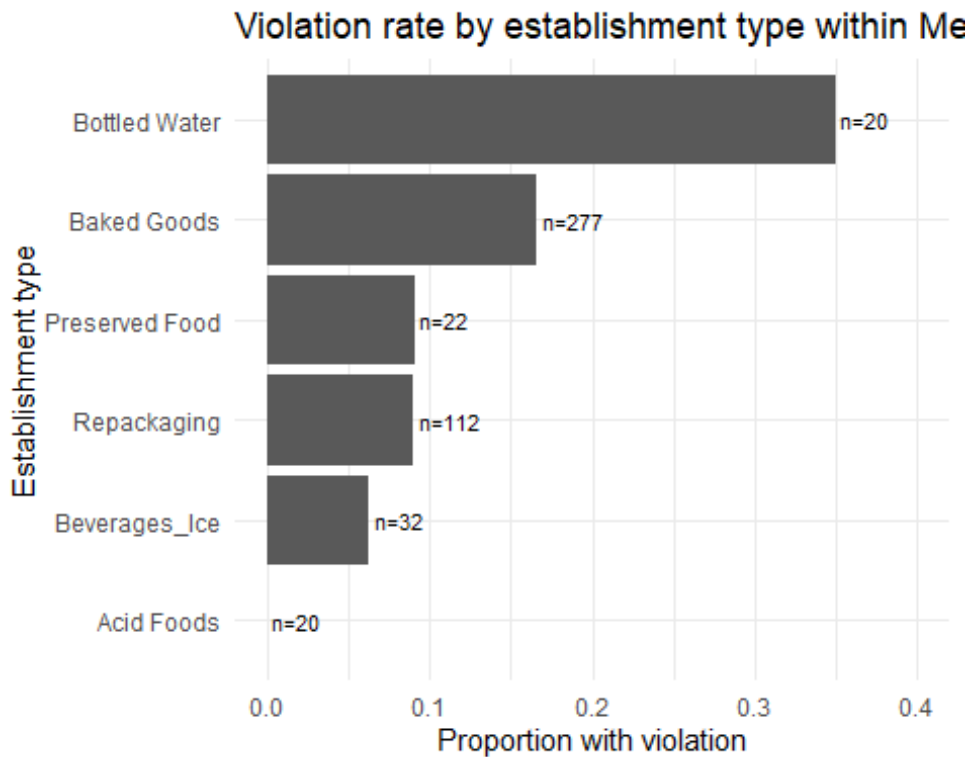
tab_med

A tibble: 6 × 4

##	est_type	n	violations	prop_violation
##	<fct>	<int>	<int>	<dbl>
## 1	Bottled Water	20	7	0.35
## 2	Baked Goods	277	46	0.166
## 3	Preserved Food	22	2	0.0909
## 4	Repackaging	112	10	0.0893
## 5	Beverages_Ice	32	2	0.0625
## 6	Acid Foods	20	0	0

```
ggplot(tab_med, aes(x = reorder(est_type, prop_violation), y = prop_violation)) +
  geom_col() +
  geom_text(aes(label = paste0("n=", n)), hjust = -0.1, size = 3) +
  coord_flip() +
  expand_limits(y = max(tab_med$prop_violation) + 0.05) +
```

```
labs(
  title = "Violation rate by establishment type within Medium-risk facilities",
  x = "Establishment type",
  y = "Proportion with violation"
) +
theme_minimal()
```



Year-over-year summary

```
tab_year <- dat %>%
  group_by(year) %>%
  summarize(
    inspections = n(),
    violations = sum(violation == "Yes"),
    pct_violations = round(100 * mean(violation == "Yes"), 1),
    .groups = "drop"
  )
```

tab_year

```
## # A tibble: 4 × 4
##   year inspections violations pct_violations
##   <fct>      <int>      <int>         <dbl>
## 1 2022         289         28           9.7
## 2 2023         388         50          12.9
## 3 2024         384         42          10.9
## 4 2025         425         41           9.6
```

```
tab_year_risk <- dat %>%
  group_by(year, risk) %>%
  summarize(
```

```

    inspections = n(),
    violations = sum(violation == "Yes"),
    pct_violations = round(100 * mean(violation == "Yes"), 1),
    .groups = "drop"
)

```

```
tab_year_risk
```

```
## # A tibble: 12 × 5
##   year risk   inspections violations pct_violations
##   <fct> <fct>         <int>         <int>         <dbl>
## 1 2022 LOW           105            2            1.9
## 2 2022 HIGH           93           13           14
## 3 2022 MEDIUM          91           13          14.3
## 4 2023 LOW          175            4            2.3
## 5 2023 HIGH          102           25          24.5
## 6 2023 MEDIUM         111           21          18.9
## 7 2024 LOW          126            5            4
## 8 2024 HIGH          120           19          15.8
## 9 2024 MEDIUM         138           18           13
## 10 2025 LOW          179            9            5
## 11 2025 HIGH          101           17          16.8
## 12 2025 MEDIUM         145           15          10.3

```

```

tab_year_risk <- tab_year_risk %>%
  mutate(
    risk = factor(risk, levels = c("HIGH", "MEDIUM", "LOW"))
  )

```

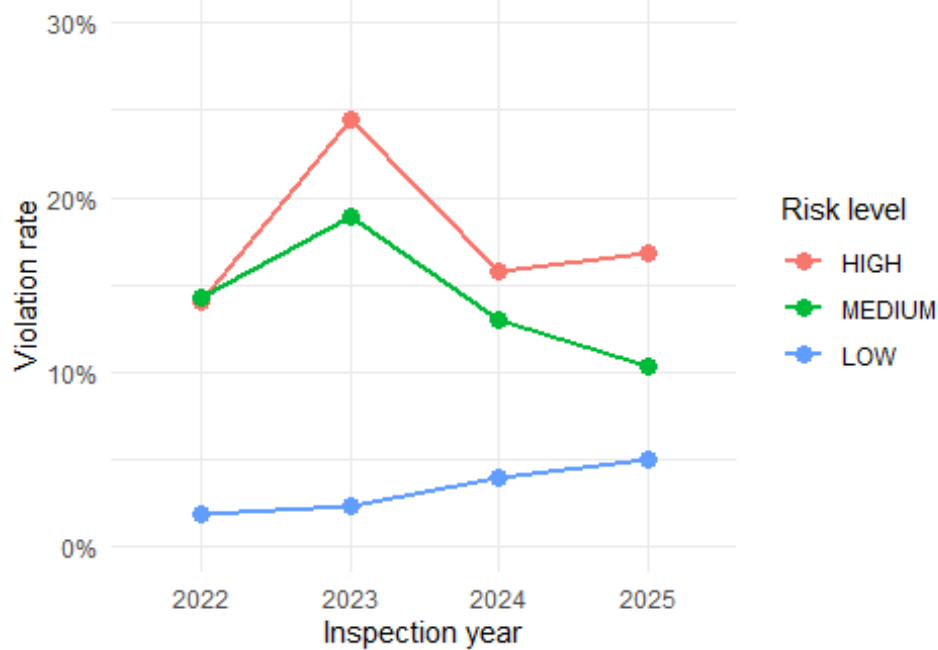
```

ggplot(tab_year_risk, aes(x = year, y = pct_violations, color = risk, group = risk
)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_y_continuous(
    limits = c(0, 30),
    labels = label_percent(scale = 1)
  ) +
  labs(
    title = "Reported violation rates were highest in 2023 among High- and Medium-
risk facilities",
    subtitle = "Annual percentage of inspections resulting in a reported violation
, by risk level",
    x = "Inspection year",
    y = "Violation rate",
    color = "Risk level"
  ) +
  theme_minimal()

```


Reported violation rates were highest in 2023 among

Annual percentage of inspections resulting in a reported violation,



```
tab_year_est_high <- dat %>%
  filter(risk == "HIGH", est_type != "0") %>%
  group_by(year, est_type) %>%
  summarize(inspections = n(), .groups = "drop") %>%
  group_by(year) %>%
  mutate(pct_of_year = round(100 * inspections / sum(inspections), 1))
```

tab_year_est_high

```
## # A tibble: 16 × 4
## # Groups:   year [4]
##   year est_type      inspections pct_of_year
##   <fct> <fct>          <int>      <dbl>
## 1 2022 Fish_Seafood      27         29
## 2 2022 Juice              6         6.5
## 3 2022 Preserved Food    24        25.8
## 4 2022 Ready_to_Eat     36        38.7
## 5 2023 Fish_Seafood     41         41
## 6 2023 Juice              1          1
## 7 2023 Preserved Food    23         23
## 8 2023 Ready_to_Eat     35         35
## 9 2024 Fish_Seafood     40        33.3
##10 2024 Juice              2         1.7
##11 2024 Preserved Food    19        15.8
##12 2024 Ready_to_Eat     59        49.2
##13 2025 Fish_Seafood     35        34.7
##14 2025 Juice              2          2
##15 2025 Preserved Food    19        18.8
##16 2025 Ready_to_Eat     45        44.6
```

```

tab_year_est_med <- dat %>%
  filter(risk == "MEDIUM", est_type != "0") %>%
  group_by(year, est_type) %>%
  summarize(inspections = n(), .groups = "drop") %>%
  group_by(year) %>%
  mutate(pct_of_year = round(100 * inspections / sum(inspections), 1))

```

```
tab_year_est_med
```

```

## # A tibble: 24 × 4
## # Groups:   year [4]
##   year est_type inspections pct_of_year
##   <fct> <fct>         <int>     <dbl>
## 1 2022 Acid Foods           2         2.2
## 2 2022 Baked Goods        53        58.2
## 3 2022 Beverages_Ice       9         9.9
## 4 2022 Bottled Water       7         7.7
## 5 2022 Preserved Food      3         3.3
## 6 2022 Repackaging        17        18.7
## 7 2023 Acid Foods          6         5.4
## 8 2023 Baked Goods        65        58.6
## 9 2023 Beverages_Ice       9         8.1
## 10 2023 Bottled Water      2         1.8
## # i 14 more rows

```